IW1RMM CW ENCODER

Firmware v7.1.4 — Software Routine Analysis

ESP32 CYD | March 2026 — IW1RMM (Mauri)

Based on original VK2IDL firmware

1. Firmware Architecture

The firmware is organised into three concurrent FreeRTOS tasks. ESP32 Core 0 runs morseTask (priority 2, stack 8192 bytes) and serialTask (priority 1, stack 4096 bytes). Core 1 runs the Arduino loop() function, which manages LVGL and the graphical interface. Communication between tasks uses a displayQueue (64 messages) and an xMutex for shared variables.

2. Initialisation Routines

Routine	Short description	Detail
setup()	Firmware entry point	Starts in sequence: SD card (before display), Serial, mutex/queue, NVS preferences, GPIO pins, LVGL, touch, HW timer, graphical UI, BLE, FreeRTOS tasks. Critical order: SD must be initialised before tft.begin() to avoid SPI bus conflicts.
setupPins()	Configure GPIO	Sets MORSE_OUT (pin 27) as output LOW, PADDLE_DAH (22) with pull-up, PADDLE_DIT (35) as input, initialises ESP32 DAC for sidetone, activates TFT backlight (pin 21).
setupLVGL()	Initialise graphics	Calls lv_init(), initialises ILI9341 display via TFT_eSPI (VSPI), configures LVGL rendering buffer (320×10 pixels), registers display and touch drivers.
setupTouch()	Initialise XPT2046 touch	Initialises dedicated HSPI bus for touch, loads calibration values from NVS namespace 'touch_cal' (if available) or uses hardcoded defaults (x: 220-3730, y: 387-3763), enables horizontal rotation.
setupTimer()	1 kHz hardware timer	Configures ESP32 hardware timer with 1 ms interrupt. Increments morseTimerTick counter used by morseDelay() for precise timing without blocking the task.
setupTasks()	Create FreeRTOS tasks	Launches morseTask on Core 0 (stack 8192, priority 2) and serialTask on Core 0 (stack 4096, priority 1). LVGL runs in loop() on Core 1.
loadPreferences()	Load NVS parameters	Reads from namespace 'cyd-morse' (NVS flash): WPM, frequency, volume, dahRatio, weight, paddleReversed, beaconInterval, contestCounter, messages msg0..msg5. Applies values and calculates Morse timing.

savePreferences()	Save NVS parameters	Writes all operating parameters plus 6 messages to NVS. Opens and closes the namespace to guarantee flash flush.
calculateMorseTiming()	Calculate Morse timing	Calculates ditLength, dahLength, elemSpace, charSpace, wordSpace in milliseconds from current WPM. If Farnsworth is active, calculates extended spacing separately.

3. Main Loop and LVGL

Routine	Short description	Detail
loop()	Main loop Core 1	Calls lv_timer_handler() to update LVGL GUI, processDisplayQueue() to apply messages from the display queue, sdPlayerTask() for SD player, checkSerialBle() for incoming BLE data, and updateClockLVGL() every second for the clock.
processDisplayQueue()	Apply display messages	Extracts messages from displayQueue produced by morseTask and updates corresponding LVGL labels (TX LED, text bars, WPM, VOL, FREQ, mode).
lvgl_updateTxLed()	Update TX LED	Changes the colour of the circular LED on the display: green when transmitting, dark grey when idle.
lvgl_appendTxBar()	Append char to text bar	Appends a character to the scrolling blue bar (L3). When L3 is full, scrolls text to rows L2 and L1.
lvgl_clearAll()	Clear display	Clears all three text bars and status labels on the display.
updateClockLVGL()	Update clock	Reads ESP32 system time and updates ui_LabelClock label every second. Label border is green if year > 1970 (RTC set), red otherwise.
my_disp_flush()	LVGL display flush	LVGL callback: transfers the rendered buffer to the ILI9341 display via tft.pushColors().
my_touchpad_read()	LVGL touch read	LVGL callback: reads raw XPT2046 coordinates, maps them to screen coordinates 320×240 via calibration, returns pressed/released state.

4. Morse Task (Core 0)

Routine	Short description	Detail
morseTask()	Main keyer task	Runs in a loop on Core 0. Waits 2 seconds at startup (FIX-4, avoids spurious 'E'). Then continuously calls checkPaddle() and sendFromBuffer(). Also manages automatic beacon and contest mode.
checkPaddle()	Read paddle state	Reads PADDLE_DIT and PADDLE_DAH pins. Based on mode (Iambic A/B, Manual, Auto)

		decides whether to start DIT or DAH transmission. Updates dit/dahPaddleFlag and decodes the Morse binary tree for display.
paddle_beginDIT()	Start DIT transmission	Activates DAC (sidetone), raises MORSE_OUT (PTT), waits DIT duration via morseDelay(), then deactivates DAC and lowers MORSE_OUT. Sends '.' symbol to display and updates decoding.
paddle_beginDAH()	Start DAH transmission	Same as paddle_beginDIT() but for DAH duration (ditLength × dahRatio).
paddle_morse_On()	Activate carrier (paddle)	Enables ESP32 DAC and raises MORSE_OUT. Used by manual key.
paddle_morse_Off()	Deactivate carrier (paddle)	Disables DAC and lowers MORSE_OUT.
paddleTimer()	Inter-element timing	Calculates timeout for current Morse character decoding. If inter-character time elapses without new elements, marks character as ready for decoding.
sendFromBuffer()	Transmit from buffer	Extracts characters from the buffer string and passes them to sendMorseChar(). Recognises <XX> prosign syntax. Updates text bar on display. Signals idle at end of buffer.
sendMorseChar(char c)	Transmit one Morse char	Converts ASCII character to DIT/DAH sequence via internal Morse table. Calls morseToneOn() and morseToneOff() for each element, respecting calculated timings. Also handles prosign characters without inter-character space.
morseToneOn(ms)	Activate tone for ms	Enables ESP32 DAC at set tone/volume, raises MORSE_OUT (PTT), waits ms via morseDelay(). Sends DISP_TX_LED ON to displayQueue.
morseToneOff(ms)	Deactivate tone for ms	Disables DAC, lowers MORSE_OUT, waits ms via morseDelay(). Sends DISP_TX_LED OFF.
morseDelay(ms)	Non-blocking delay	Waits ms milliseconds using morseTimerTick counter (1 kHz HW timer). Yields CPU every 1 ms with vTaskDelay(1) instead of blocking delay().
onMorseTimer()	HW timer ISR	IRAM_ATTR interrupt service routine: increments morseTimerTick every millisecond.
addToBuffer(str)	Add text to buffer	Appends string str to the transmission buffer, adding a separator space. Thread-safe via LOCK/UNLOCK.
clearBuffer()	Clear buffer	Clears the transmission buffer and sends DISP_CLEAR to the display queue.
sendProsign(str)	Transmit prosign <XX>	Transmits prosign characters without inter-character space between them, as per international telegraph standard.

5. Serial Task (Core 0)

Routine	Short description	Detail

serialTask()	Serial task Core 0	Continuously reads USB serial port (Serial). In K3NG mode calls checkSerial(); in WinKey mode calls checkSerialWinkey(). Low priority (1) to avoid interfering with the keyer.
checkSerial()	K3NG command parser	Interprets K3NG commands arriving from serial port. Commands start with '\' followed by a letter. Handles: \W (set WPM), \F (set frequency), \Z (set time), \A-\Z (memories), \J (reset touch calibration), \Y (sidetone freq), \X (HSCW toggle), \E (extended display), and all configuration commands. Normal text is queued to the transmission buffer.
checkSerialWinkey()	WinKey protocol parser	Interprets the WinKey 2.x binary protocol. Handles host open (0x00), speed commands (0x02), mode register (0x0E), text (0x04), tune (0x0B), PTT timing, and all commands defined in the WK3 datasheet.
checkSerialBle()	BLE command parser	Processes data received in bleRxBuffer. Applies the same parser as checkSerial() or checkSerialWinkey() depending on the active mode.
sendStatus()	Send serial status	Responds to \S command by sending current state to serial port: WPM, freq, vol, mode, beacon, BLE, SD.
sendHelp()	Send serial help	Prints the list of available commands to serial port.
sendContestMsg()	Transmit contest message	Queues contest message with progressive counter to transmission buffer, then increments counter.

6. BLE (Bluetooth Low Energy)

Routine	Short description	Detail
bleSetup()	Initialise BLE	Initialises BLEDevice with name 'VK2IDL_Morse', creates BLE server and Nordic UART service (NUS) with TX and RX characteristics. Registers MyServerCallbacks and MyCallbacks.
bleInit()	Start BLE advertising	Makes the device visible to BLE clients. Sets device name explicitly in primary advertising payload (fix v7.1.3 — resolves 'unnamed' on Android). Moves service UUID to scan response. Updates BLE button on screen.
bleStop()	Stop BLE	Stops advertising and disconnects clients. Releases BLE resources.
bleSend(str)	Send string via BLE	Sends string str to connected BLE client via TX characteristic. Does nothing if no client is connected or BLE is disabled.
bleSendLong(str)	Send long string via BLE	Sends long strings in chunked 20-byte packets with inter-packet delay. Used for \H, \V, \I, \P list responses.
updateBleButton()	Update BLE UI button	Updates BLE button text and colour in Settings tab based on state: OFF / ADV / CONN.

MyServerCallbacks	BLE connection callback	C++ class: onConnect() sets bleClientConnected=true and signals UI update. onDisconnect() restarts advertising if BLE is enabled.
MyCallbacks	BLE receive callback	C++ class: onWrite() queues received bytes in bleRxBuffer for processing in loop().

7. LVGL User Interface

Routine	Short description	Detail
createUI()	Create main tabview	Creates tabview with 5 tabs: Morse 1, Morse 2, SD, Settings, Spare. Calls creation function for each tab.
createTabMorse1()	Create Morse 1 tab	Builds main tab layout: scrolling text areas L1/L2/L3, message buttons (CQ, NAME, ANT, QTH, RST, TEST), CLEAR, STOP, TUNE, BEACON, CONTEST, MACRO, KEYER MODE buttons, WPM/VOL/FREQ controls.
createTabMorse2()	Create Morse 2 tab	Alternative keyer layout with same functionality but optimised arrangement for operating use.
createTabSD()	Create SD player tab	Builds SD panel: file/info/text labels, OPEN/PLAY/PAUSE/STOP buttons, WPM label. Updates labels with current SD card state.
createTabSettings()	Create Settings tab	Builds Settings panel: Ratio, Weight, Pad Reverse, Beacon, Contest, Farnsworth, SD Log, QRSS, BLE, Protocol (K3NG/WinKey), Recalibrate buttons.
createTabSpare()	Create Spare tab	Development tab with HSCW (High Speed CW) functionality and future controls.
createCommonControls()	Create common controls	Creates WPM+/-, VOL+/-, FREQ+/- controls, TX LED, Iambic Mode button — present identically in both Morse 1 and Morse 2 tabs.
createLogAreas()	Create scrolling text areas	Creates three LVGL labels L1, L2, L3 (Morse text bars) on a parent panel. L3 is the active blue bar (large font), L2 and L1 are black (small font).

8. Button Event Handlers

Handler	Trigger	Action
onCQ/Name/AntRig/Qth/Rst/Test()	Tap on message button	Adds preset message to buffer. Long-press opens showMsgEditor() for editing.
onMacro()	Tap MACRO	Adds macroMessage to buffer. Long-press opens editor to modify macro.
onStop()	Tap STOP	Sets stopRequested=true. morseTask interrupts transmission at end of current element

		and clears buffer.
onClear()	Tap CLEAR	Calls clearBuffer() and sends DISP_CLEAR to display queue.
onTune()	Tap TUNE	Toggles TUNE mode: sends continuous carrier + PTT ON. Updates UI.
onBeacon()	Tap BEACON	Toggles automatic CQ message beacon transmission at set interval.
onContest()	Tap CONTEST	Opens Contest popup to set starting counter number. Toggles contest mode.
onKeyerMode()	Tap KEYER/BUFFER	Toggles between KEYER mode (transmits immediately from paddle) and BUFFER mode (accumulates in buffer before transmitting).
onIambicMode()	Tap iambic mode	Cycles between modes: Iambic A → Iambic B → Manual → Auto. Updates label and saves preference.
onWpmUp/Down()	Tap +/- WPM	Increments/decrements WPM by 1 (range 5-100). Recalculates Morse timing and updates display.
onVolUp/Down()	Tap +/- VOL	Changes DAC attenuation level (0=mute, 4=max). Updates display and hardware DAC.
onFreqUp/Down()	Tap +/- FREQ	Modifies sidetone frequency by 50 Hz (range 300-1200 Hz). Updates DAC.
onRatio()	Tap RATIO	Opens showPopupRatio(): slider to modify dahRatio (2.0-4.5).
onWeight()	Tap WEIGHT	Opens showPopupWeight(): slider to modify spacing_Weight.
onPadRev()	Tap PAD REV	Opens showPopupPadRev(): toggle to swap DIT/DAH on paddle.
onFarnsworthToggle()	Tap Farnsworth	Toggles Farnsworth. If activated, opens showPopupFarnsworth() to set spacing WPM.
onQrssToggle()	Tap QRSS	Cycles QRSS modes: OFF → 3s → 6s → 10s → 30s per DIT. Recalculates qrssDitMs.
onBleToggle()	Tap BLE	Toggles Bluetooth Low Energy. Calls bleInit() or bleStop().
onProtocolToggle()	Tap K3NG/WinKey	Toggles serial protocol between K3NG (115200 baud) and WinKey 2.x (1200 baud). Updates winkeyMode and reinitialises Serial at correct baud rate.
onSdLogToggle()	Tap SD Log	Toggles SD logging. If activated, calls sdLogInit() to open/create log file.
onRecalibrate()	Tap Recalibrate	Clears touch calibration from NVS and reboots device to start guided calibration procedure.
showMsgEditor(idx)	Long-press on message	Opens overlay with LVGL textarea pre-loaded with current message and virtual keyboard. SAVE and CANCEL buttons close the editor.
onMsgEditorSave()	Tap SAVE in editor	Reads text from textarea, updates global message variable, performs NVS flush with end()/begin()/putString()/end()/begin() to guarantee persistence.

9. Popups and Dialogs

Routine	Short description	Detail
createPopupBase(title)	Create popup structure	Creates a semi-transparent centred modal panel with title. Used as base by all popups.
closePopup()	Close active popup	Deletes popupBox and resets all current popup widget references.
showPopupRatio()	DAH Ratio popup	Shows slider to adjust dahRatio with -/+ buttons, current value label, SAVE button.
showPopupWeight()	Spacing Weight popup	Same as showPopupRatio() but for spacing_Weight.
showPopupBeacon()	Beacon Interval popup	Allows setting beacon interval in seconds (10-600).
showPopupContest()	Contest Counter popup	Allows setting the contest counter starting number.
showPopupFarnsworth()	Farnsworth WPM popup	Shows two labels (Effective WPM and Chars WPM) and slider for farnsworthWPM. Values update in real time while moving the slider.
showPopupHscw(wpm)	HSCW Speed popup	Slider to set HSCW (High Speed CW) speed. Called from Spare tab.

10. SD Card and Player

Routine	Short description	Detail
sdScanFiles(target)	Scan SD for .txt files	Reads SD card root directory and populates sdFileList[] with found .txt file names (max 20). If target is specified, repositions cursor on that file.
sdLoadFile(filename)	Load file from SD	Opens specified .txt file, reads all content into sdText, resets sdCharIndex=0 and updates text preview labels.
sdOpenFilePopup()	Open file list popup	Creates popup with scrollable list of available .txt files on SD. Tap on a file calls sdLoadFile().
sdCloseFilePopup()	Close file list popup	Deletes sdPopup LVGL object.
sdPlayerTask()	SD player task (in loop)	Called every loop() cycle. If sdState==SD_PLAYING, extracts next character from sdText and adds it to Morse transmission buffer, advancing sdCharIndex. When end of file reached, sets sdState to SD_IDLE.
sdUpdateFileLabel()	Update SD file label	Shows current file name and text length on SD tab.
sdUpdateInfoLabel()	Update SD info label	Shows current frequency and volume in SD tab.

sdUpdateWpmLabel()	Update SD WPM label	Updates WPM label in SD tab with current value.
sdUpdateButtons()	Update SD button state	Enables/disables PLAY/PAUSE/STOP buttons based on sdState (IDLE/PLAYING/PAUSED).
sdLogInit()	Initialise SD log	Creates or opens a CSV log file on SD with name based on uptime. Writes header with date, WPM, frequency.
getUptimeString()	Return uptime string	Calculates and returns time elapsed since boot in HH:MM:SS format.

11. Touch Calibration (NEW v7.1.4)

Routine	Short description	Detail
runTouchCalibration()	Run guided calibration	Runs the 4-point interactive calibration procedure. Displays crosshair at each corner in sequence. Waits for user touch via waitTouchPoint(). Calculates and validates the 4 calibration values. Shows ACCEPT/RETRY screen. If accepted (or timeout 10s), saves to NVS namespace 'touch_cal'.
waitTouchPoint(x,y,label)	Wait for touch on point	Displays crosshair at coordinates (x,y) with on-screen instruction label. Waits for a touch event from XPT2046 with timeout. Serial 'R' escape cancels. Returns raw ADC coordinates.
loadTouchCalibration()	Load calibration from NVS	Reads cal_x_min, cal_x_max, cal_y_min, cal_y_max from 'touch_cal' NVS namespace. Validates range (x_min 100-1000, x_max 3000-4095, excursion ≥ 2000). If valid, applies to touch mapping. If invalid or absent, uses hardcoded defaults.
saveTouchCalibration()	Save calibration to NVS	Writes the 4 calibration values to 'touch_cal' NVS namespace with putInt(). Opens and closes namespace to guarantee flash flush.
resetTouchCalibration()	Reset calibration (\J)	Clears 'touch_cal' NVS namespace and triggers device reboot to restart guided calibration. Called by K3NG \J command and onRecalibrate() handler.

12. Utility Functions

Routine	Short description	Detail
postDisplay(type, val)	Send message to display queue	Thread-safe inline: builds a DisplayMsg and sends it to displayQueue without blocking. If queue is full, message is discarded.
postDisplayChar(c)	Send char to text bar	Specialisation of postDisplay() for type DISP_TX_BAR: sends a single character to the blue text bar.
updateStatusLabels	Update status labels	Updates WPM, VOL, FREQ labels in both Morse 1 and Morse 2 tabs.

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getModeName()	Return mode name	Returns current keyer mode string: 'IAMB-A', 'IAMB-B', 'MANUAL', 'AUTO'.
calculateDitDuration()	Calculate DIT duration ms	Returns DIT duration in milliseconds at current WPM: 1200/WPM.
volumeUiLevel(v)	Convert volume UI level	Inline: converts internal DAC level (0-4) to screen display value (inverted: 0=mute, 4=max).
getQrssModeString(mode)	QRSS mode string	Returns text representation of active QRSS mode: 'OFF', '3s', '6s', '10s', '30s'.